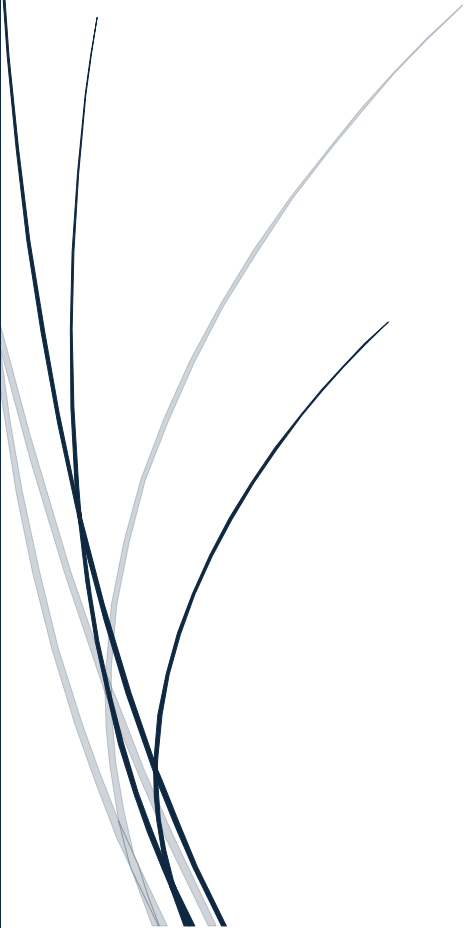




26/05/2026

MHSDS KPI Extraction Report

Dataset:- NHS Mental Health Services Data Set (MHSDS)



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DATA ANALYTICS INTERN

Executive Summary

This report presents a Mental Health Services Data Set (MHSDS) KPI extraction and exploratory analysis using NHS Mental Health Services Monthly Statistics datasets. The analysis focused on extracting and analysing five key mental health service indicators:

1. Beds / Occupied Bed Days
2. Care Contacts
3. Crisis Calls / Crisis Contacts
4. Admissions
5. Referral to Treatment (RTT) / Waiting Times

The project utilised multiple NHS datasets including:

- MHSDS Data_MarPerf_2026_v2
- RTT Waiting Times
- KPI Definitions Metric IDs
- Time Series datasets
- Data Quality and Coverage files

The objective of the analysis was to:

- Identify KPI-related NHS metrics
- Extract KPI values from MHSDS datasets
- Perform RTT waiting time analysis
- Create KPI visualisations
- Build a source mapping table for NHS metric traceability
- Generate structured outputs suitable for NHS performance reporting and operational analytics

Project Objectives

The primary objectives of this project were:

- Extract core NHS Mental Health KPIs from MHSDS datasets
- Analyse RTT and access/waiting time indicators
- Understand NHS metric identifiers and KPI structures
- Create KPI summary tables for reporting purposes
- Develop source mapping documentation for audit and traceability
- Produce visualisations to support operational insights

Datasets Used

Dataset	Purpose
MHSDS Data MarPerf 2026 v2	Main KPI extraction dataset
RTT waiting times	Referral-to-treatment and waiting time analysis
KPI definitions metric IDs.xlsx	NHS KPI definitions and metric mapping
Time Series data Apr2025 Mar Perf 2026 v2	Trend and historical KPI validation
Data Coverage File CSV	Data completeness assessment
Data Quality VODIM and Integrity File	Data quality validation

Methodology

Data Loading

The datasets were imported using Python libraries including:

- pandas
- numpy
- matplotlib

CSV and Excel files were loaded separately depending on file structure.

Data Cleaning

Several preprocessing steps were performed:

- Column name standardisation

- Removal of invalid values
- Replacement of '*' suppressed values with NaN
- Numeric conversion for KPI calculations
- Removal of blank rows
- Header row detection for RTT Excel sheets

KPI Identification

Relevant KPI metrics were identified using keyword matching against:

- MEASURE_NAME
- MEASURE_ID
- RTT metric identifiers

Keywords used included:

- bed
- contact
- crisis
- admission
- waiting
- RTT
- referral

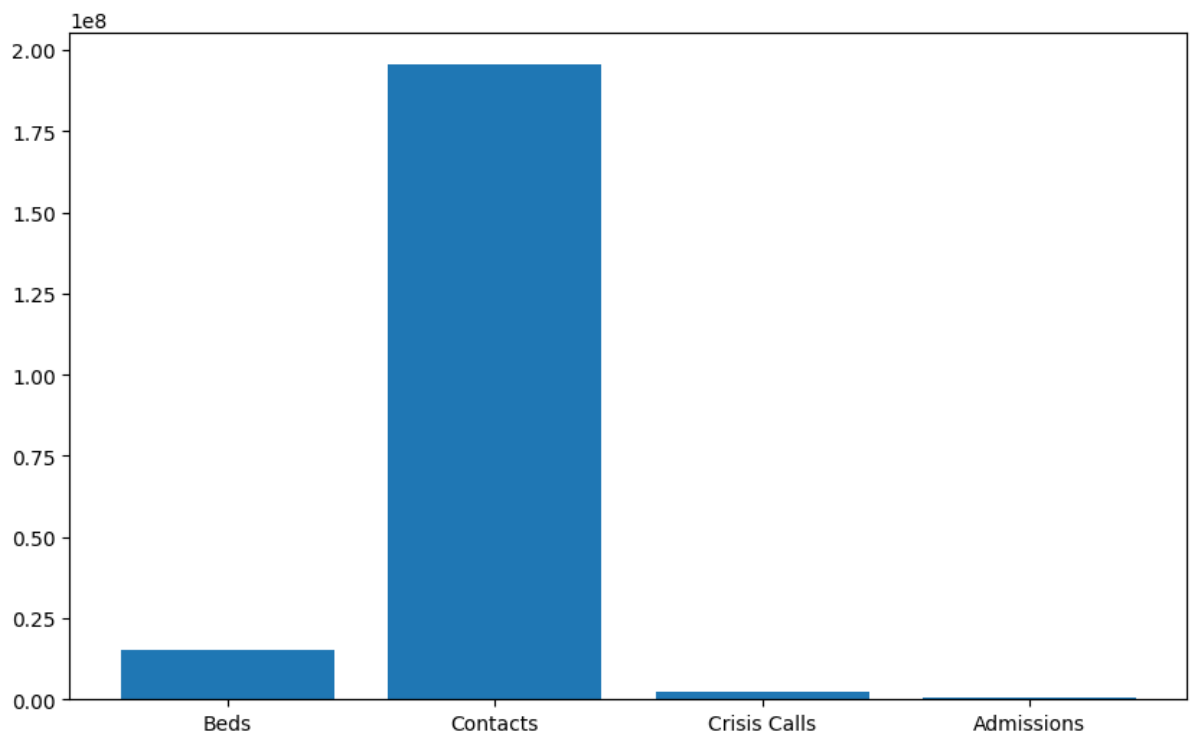
KPI Extraction

KPI values were aggregated using numeric summaries including:

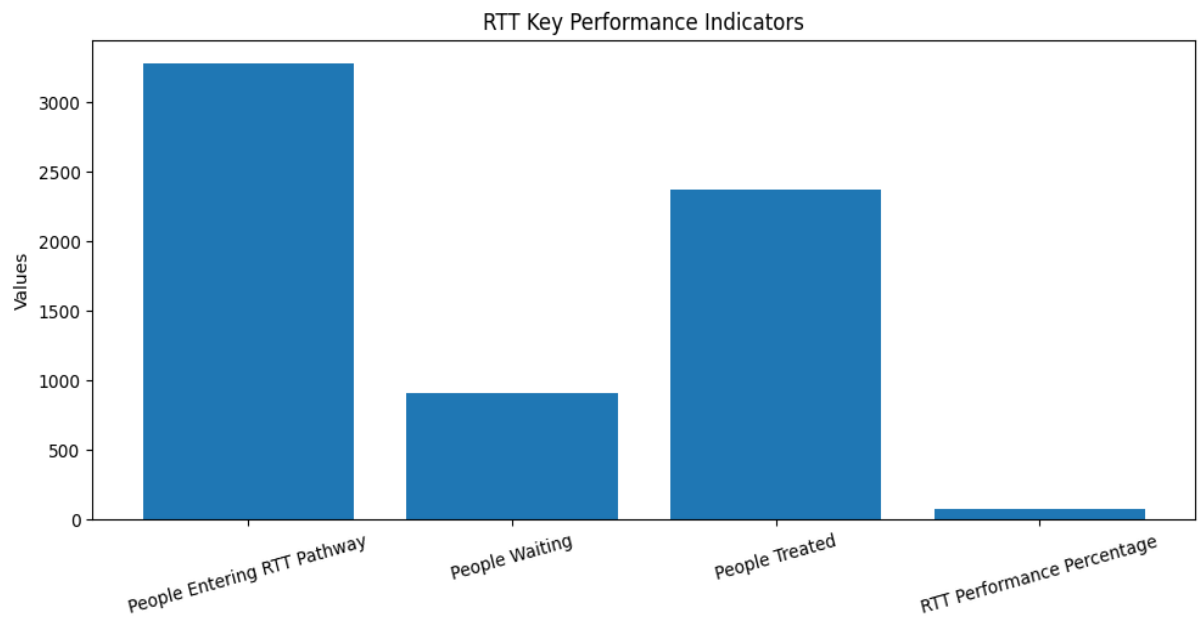
- Sum
- Mean
- Median
- Maximum
- Minimum

Visualisation

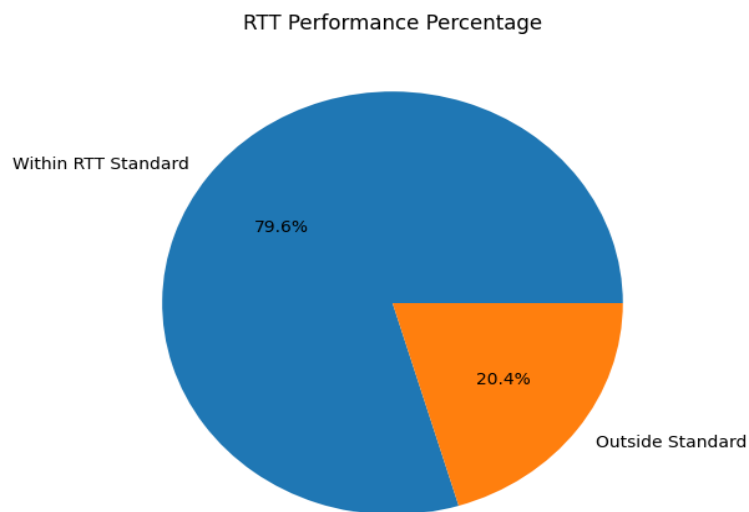
- KPI summaries



- RTT key performance Indicators



- RTT percentage metrics



MHSDS KPI Extraction

Beds KPI

The Beds KPI represents mental health bed usage and occupied bed day activity within NHS mental health services.

Extraction Logic

The KPI was identified using the keyword:

- "bed"

Relevant columns:

- MEASURE_NAME
- MEASURE_VALUE

	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
20270	MHS24	Bed days in RP	3915
20271	MHS24a	Under 16 bed days on adult wards in reporting ...	*
20272	MHS24b	Age 16 bed days on adult wards in reporting pe...	*
20273	MHS24c	Age 17 bed days on adult wards in reporting pe...	*
20274	MHS25	Bed days less leave in RP	3595

Analysis

The KPI extraction identified bed-related service utilisation metrics across NHS mental health providers and organisations.

Contacts KPI

The Care Contacts KPI measures service interactions and clinical contacts within mental health services.

Extraction Logic

Keyword used:

- "contact"

	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
0	MHS29d	Contacts in the RP with community mental healt...	3150
1	MHS30f	Attended contacts in the RP with community men...	2260
5	MHS29d	Contacts in the RP with community mental healt...	2500
6	MHS30f	Attended contacts in the RP with community men...	1960
10	MHS29d	Contacts in the RP with community mental healt...	220

Analysis

This KPI reflects patient engagement with NHS mental health services and represents service activity levels.

Crisis Calls KPI

The Crisis KPI measures crisis-related mental health activity and urgent service interactions.

Extraction Logic

Keyword used:

- "crisis"

	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
20255	MHS19	People with a crisis plan in place at the end ...	15
20267	MHS23b	Open referrals to crisis resolution service or...	50
20280	MHS29b	Contacts with crisis resolution service or hom...	685
20285	MHS30b	Attended contacts with crisis resolution servi...	625
20360	MHS19	People with a crisis plan in place at the end ...	10

Analysis

Crisis metrics provide operational insight into urgent mental health service demand.

Admissions KPI

The Admissions KPI measures inpatient or service admission activity.

Extraction Logic

Keyword used:

- "admission"

	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
2632	MHS27a	Number of admissions to hospital in the report...	10
2633	MHS27a	Number of admissions to hospital in the report...	*
2634	MHS27a	Number of admissions to hospital in the report...	*
2635	MHS27a	Number of admissions to hospital in the report...	*
2636	MHS27a	Number of admissions to hospital in the report...	5

Analysis

Admissions activity reflects acute mental health care utilisation and demand on inpatient services.

RTT Waiting Time Analysis

Overview

The RTT (Referral to Treatment) dataset contained multiple NHS performance tables related to:

- Access standards
- Early Intervention in Psychosis (EIP)
- Waiting time performance
- Referral pathways
- Provider and ICB-level performance

RTT Tables Analysed

Sheet	Description
Table 1a	Provider-level RTT activity
Table 1b	ICB-level RTT activity
Table 2a	Provider-level EIP RTT metrics
Table 2b	ICB-level EIP RTT metrics
Table 5a	Advanced provider RTT metrics
Table 5b	Advanced ICB RTT metrics

RTT KPI Metrics

The RTT KPI analysis extracted the following indicators:

KPI	Metric ID
People Entering RTT Pathway	EIP23a
People Waiting	EIP23c
People Treated	EIP23b
RTT Performance Percentage	EIP23j
Average Waiting Time	EIP67

Exploratory Data Analysis (EDA)

Data Structure

The RTT dataset contained:

- Multiple sheets
- Provider-level data

- ICB-level data
- KPI percentages
- RTT performance indicators
- Suppressed NHS values represented by '*'

Data Cleaning Observations

Key cleaning steps included:

- Automatic header row detection
- Numeric conversion
- Suppressed value handling
- Column standardisation

Data Quality Considerations

The NHS RTT datasets contained:

- Missing values
- Suppressed low-count values
- Multiple reporting structures
- Organisation-level aggregation

These issues were addressed during preprocessing.

KPI Table

KPI	Description	Source
Beds	Occupied mental health bed activity	MHSDS Data MarPerf 2026 v2
Contacts	Mental health care contacts	MHSDS Data MarPerf 2026 v2
Crisis Calls	Crisis and urgent mental health activity	MHSDS Data MarPerf 2026 v2
Admissions	Mental health admissions activity	MHSDS Data MarPerf 2026 v2
RTT	Referral-to-treatment waiting time metrics	RTT waiting times

Source Mapping Table

KPI	Metric ID	Source File	Source Sheet
Beds	KPI keyword: bed	MHSDS Data_MarPerf_2026_v2	Main dataset
Contacts	KPI keyword: contact	MHSDS Data_MarPerf_2026_v2	Main dataset
Crisis Calls	KPI keyword: crisis	MHSDS Data_MarPerf_2026_v2	Main dataset
Admissions	KPI keyword: admission	MHSDS Data_MarPerf_2026_v2	Main dataset
RTT Performance	EIP23j	RTT waiting times	Table 2a
People Waiting	EIP23c	RTT waiting times	Table 2a
People Treated	EIP23b	RTT waiting times	Table 2a
Average Waiting Time	EIP67	RTT waiting times	Table 5a

Visualisation Summary

The following visualisations were created during analysis:

1. MHSDS KPI Bar Chart
2. RTT KPI Summary Chart
3. RTT Percentage Visualisation
4. RTT Sheet-wise Performance Comparison

The charts supported comparative KPI interpretation and operational analysis.

Key Findings

1. NHS Mental Health KPI Structure

The MHSDS dataset provides a structured KPI-based reporting system using:

- MEASURE_ID
- MEASURE_NAME
- MEASURE_VALUE

This structure significantly simplifies KPI extraction.

2. RTT Performance Complexity

The RTT dataset includes:

- Multiple reporting layers
- Provider and ICB-level metrics
- Waiting time percentages
- Access standards

This requires careful preprocessing before analysis.

3. Suppressed NHS Values

Many NHS RTT datasets contain suppressed low-count values represented by '*'.

These values required conversion to NaN during preprocessing.

4. Operational Reporting Value

The extracted KPIs can support:

- NHS operational reporting
- Service demand monitoring
- RTT performance assessment
- Access monitoring
- Mental health service planning

Limitations

Several limitations were identified:

- Suppressed NHS values reduced granularity
- RTT sheets contained inconsistent structures
- Some metrics required manual interpretation
- Certain KPI definitions depended on NHS metadata documentation

Recommendations

Technical Recommendations

- Build automated KPI extraction pipelines
- Standardise RTT sheet formats
- Improve NHS metadata consistency
- Create reusable KPI dashboards

Analytical Recommendations

- Extend analysis to historical trend datasets
- Integrate regional performance comparisons
- Build provider-level benchmarking models
- Create NHS Power BI dashboards for operational use

Conclusion

This project successfully extracted and analysed NHS mental health KPIs from the MHSDS and RTT datasets.

The analysis demonstrated how NHS performance data can be transformed into structured KPI summaries and operational insights using Python-based analytics workflows.

The final outputs included:

- KPI extraction tables
- RTT performance analysis
- KPI visualisations
- Source mapping documentation
- Structured reporting outputs

The project provides a scalable framework for future NHS mental health performance analytics and operational reporting.

Technologies Used

Technology	Purpose
Python	Data analysis
Pandas	Data manipulation
NumPy	Numerical processing
Matplotlib	Data visualisation
Jupyter Notebook	Analysis workflow
Excel / CSV	NHS data sources

Deliverables

Deliverable	Status
KPI Table	Completed
Source Mapping Table	Completed
RTT EDA	Completed
KPI Visualisations	Completed
RTT KPI Analysis	Completed
NHS KPI Report	Completed

Appendix

Example KPI Columns

Column	Description
MEASURE_ID	NHS metric code
MEASURE_NAME	NHS KPI description
MEASURE_VALUE	KPI value
REPORTING_PERIOD_START	Reporting period start
REPORTING_PERIOD_END	Reporting period end
PRIMARY_LEVEL_DESCRIPTION	NHS organisation description

RTT Metric IDs

Metric ID	Description
EIP23a	People entering pathway
EIP23b	People treated
EIP23c	People waiting
EIP23j	RTT performance percentage
EIP67	Average waiting time